

PRINCIPAL AI ARCHITECTURE

LONGITUDINAL CLINICAL INTELLIGENCE & CARE COORDINATION PLATFORM

GOVERNED AGENTIC AI OPERATING MODEL

Principal AI Architecture for longitudinal state computation, bounded multi-agent orchestration, hybrid retrieval, predictive models, speech intelligence, interoperability, resilience, decision lineage, trust engineering and outcome-governed autonomy.

ARCHITECTURAL POSITION

Deterministic workflow spine · bounded probabilistic reasoning · policy-mediated tools
Immutable decision lineage · explicit abstention · accountable humans retain authority

LONGITUDINAL STATE · AGENTIC AI · CLINICAL OPERATIONS · CONTROL ENGINEERING

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The document combines an end-to-end technical architecture with a Principal-level decision-engineering compendium. Every consequential choice states why it exists, where it is enforced, when it applies, how it fails, and what evidence permits promotion.

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1 Problem Statement

PRINCIPAL ARCHITECTURE POSITION

The platform is not a chatbot or workflow wrapper. It is a versioned decision system in which longitudinal context, evidence, policy, model outputs, human authority and external effects are reconstructable as one governed chain.

1.1 The Longitudinal Operational Gap

Patient-facing and staff-facing work is fragmented across calls, messages, schedules, referral documents, discharge events, EHR records, care-gap files and manual queues. These sources disagree on identity, event time, authority and completion. A ticketing layer can route tasks, but cannot determine whether an action remains applicable, whether consent permits it, whether an event supersedes an earlier decision, or whether an external write actually committed.

The architectural unit is therefore the CareCoordinationDecisionRecord, not a model turn. It binds the subject, episode, as-of time, source facts, retrieved evidence, feature vector, policy result, exact artifacts, model proposal, validation defects, human decision, tool request, external receipt, reconciliation state and eventual outcome.

1.2 Why This Is an AI Systems Problem

Failure of rules-only automation	AI systems consequence	Hard control
Unstructured multilingual input	Speech recognition, extraction, entity resolution and grounded generation form an evaluated portfolio.	Every extracted fact carries source pointer, confidence, temporal scope and validation.
Incomplete and out-of-order longitudinal state	Temporal feature computation and correction are required.	Event time, availability time and processing time remain distinct.
Actions have asymmetric risk	Models may recommend or draft, but deterministic policy owns permission.	Identity, consent, safety, eligibility and side effects are not prompt decisions.
Outcome labels mature late	Training needs exposure logging, point-in-time datasets and maturity windows.	Future data and unexposed candidates cannot become labels.
Evidence conflicts	Hybrid retrieval and bounded graph traversal surface support and contradiction.	Retrieval score is not truth; unsupported claims abstain.

1.3 Problem Definition

Build a multi-tenant Azure platform that constructs a replayable longitudinal patient state, detects executable coordination work, ranks admissible actions, invokes only authorized low-risk side effects, and captures a complete audit and learning trail. It must support real-time voice, asynchronous multi-day workflows, batch outreach, human review, EHR reconciliation, regional continuity, artifact rollback and tenant-specific governance without platform forks.

1.4 Non-Negotiable Constraints

- No foundation model owns canonical clinical, consent, identity, safety or workflow state.
- The EHR remains authoritative for clinical documentation; the platform owns coordination workflow and reconciliation.
- Every external effect is typed, idempotent, attributable, policy-authorized and recoverable from an uncertain commit.
- Autonomy is earned per tenant, intent, language and action class, and can automatically decrease.
- Models, prompts, agents, schemas, features, policies, indexes and tools are independently versioned and rollback-capable.
- Patient and tenant isolation is enforced in identity, network, store, index, cache, feature and telemetry layers—not in prompts.

2 Project Overview

PRINCIPAL ARCHITECTURE POSITION

Six bounded coordination products share a common AI control plane, durable workflow spine, longitudinal state, feature platform, knowledge services and governance plane. Shared intelligence does not imply shared mutable state.

2.1 Platform Mission

The target platform converts fragmented observations into coordinated, explainable and recoverable operational action. It delivers Call, Visit, Care Gap, Transitions, Referral and Care Management capabilities without allowing any agent to become the workflow system of record.

2.2 Bounded Contexts

Context	Primary aggregate	AI contribution	Authority
Call	Session, utterance, intent, disposition	Streaming ASR, language ID, extraction, grounded response and summary	Telephony session plus disposition policy
Visit	Slot, hold, appointment, confirmation	Constraint-aware ranking and conversational clarification	Scheduling service and EHR reconciliation
Care Gap	Measure instance, evidence status, outreach	Evidence extraction, prioritization and response classification	Measure engine and source evidence
Transitions	Admission/discharge episode and follow-up	Event correlation, segmentation and next-action proposal	Transition state machine and human reviewer
Referral	Packet, checklist, destination and closure	Extraction, requirement analysis and destination ranking	Referral workflow and acknowledgement
Care Management	Episode, tasks, goals and touchpoints	Longitudinal summary, task prioritization and draft outreach	Human-owned plan with constrained automation

2.3 Platform Shape

Plane	Needs	Azure implementation	Invariant
Edge	Channel ingress and protection	Front Door Premium, WAF, API Management, Communication Services	Identity, consent and admission precede processing.
Workflow	Timers, retries, waits and compensations	Durable Functions, Service Bus, Event Grid	Workflow history is authoritative.
Agentic	Context, bounded planning and tool selection	Microsoft Foundry Agent Service and model routes	Agents propose; policy and tools authorize.
State	Aggregates, idempotency, outbox and projections	Azure Database for PostgreSQL, Cosmos DB, Redis	One writer per aggregate; versioned mutation.
Knowledge/feature	FHIR, corpus, graph and temporal features	AHDS, Azure AI Search, ADLS Gen2, Cosmos DB	Every projection rebuilds from canonical sources.
Model/eval	Training, registry, endpoints and scorecards	Azure Machine Learning, MLflow, Foundry evaluations	Promotion requires linked evidence and rollback.
Security/SRE	Identity, secrets, network, operations	Entra ID, Managed Identity, Key Vault, Private Link, Sentinel	No agent-held secret; every PHI path observable.

2.4 Capacity Posture

Capacity is derived from measured arrival distributions, p95 service time, failover headroom, token distributions, concurrent speech streams, active workflow count, message size and external rate limits. Numbers are never invented in the architecture document. Each tenant receives a load profile, quota model, dependency budget and overload behavior before onboarding.

3 Product Capability Baseline

Capability	Required behavior	Architecture implication	Release evidence
Omnichannel intake	Correlate interactions into one episode.	Canonical envelope, identity confidence, dedup key and raw pointer.	Correlation precision/recall and duplicate rate.
Longitudinal state	Expose current state and temporal history.	Bitemporal observations, episode boundaries and projection watermark.	Replay parity and late-event convergence.
Next-best action	Rank eligible actions under policy and capacity.	Deterministic eligibility then calibrated ranking with no-action class.	Calibration, uplift, abstention and subgroup error.
Grounded assistance	Answer from approved patient and policy evidence.	ACL pre-filtered hybrid retrieval and citation map.	Evidence recall, entailment and unsupported-claim rate.
Human review	Interrupt before high-risk or uncertain action.	Durable callback, artifact hash, role, expiry and revalidation.	Review SLA, override and stale-approval rate.
Closed-loop integration	Write and reconcile EHR/partner outcomes.	Saga, idempotency, inbox/outbox and uncertain-commit resolver.	Backlog age and duplicate external effect.
Learning loop	Convert outcomes and edits into labels.	Exposure log, maturity window and point-in-time compiler.	Label coverage, delay and leakage tests.

4 Real-Time and Operational Constraints

PRINCIPAL ARCHITECTURE POSITION

Latency, throughput and recovery are modeled as decision-path properties. A global p95 hides the exact stage that violated the conversational, safety or external-commit contract.

4.1 Quality Attribute Scenarios

Stimulus	Expected response	Acceptance measure
Inbound burst during active campaign	Voice admission remains protected; nonurgent work queues by priority.	No silent loss; voice budget preserved; bounded queue age.
Model deployment throttles	Router selects an evaluated fallback or deterministic template.	No untested model path; route reason retained.
EHR times out after mutation	Mark COMMIT_UNCERTAIN and query by idempotency key before retry.	Zero duplicate appointment or referral mutation.
Late discharge event	Insert by event time, recompute affected window and cancel stale outreach.	Projection converges and superseded action cannot execute.
Prompt injection in referral PDF	Treat as untrusted data; no tool authority; validate output and destination.	No privilege expansion or exfiltration.
Regional outage	Fence old write epoch, activate paired cell and reconcile workflows.	Tier-specific RTO/RPO and no split brain.

4.2 Planning Load Model

For workload class c , required concurrency is $N_c = \lambda_c \times S_c \times (1 + H_c)$, where λ is peak arrival rate, S is p95 service time and H is retry, burst and failover headroom. Model demand is planned in requests and input/output tokens; speech in concurrent streams and real-time factor; durable workflow in active instances, timers, history growth and queue partitions.

4.3 Voice Latency Budget

Stage	Budget intent	Degradation before deadline
Edge/auth	Verify tenant, consent and session claims.	Reject or live-transfer; never skip identity control.
ASR stable partial	Generate tentative intent without side effects.	Ask for repetition or switch to callback.
Safety/intent	Run independent compact classifier and rules.	Escalate on positive or uncertain high-severity signal.
Context fetch	Acquire minimum exact and retrieved evidence.	Drop optional retrieval, not required policy or identity facts.
Generation/TTS	Create validated response and first audio.	Use approved compact route, then deterministic template.

4.4 Operational Semantics

- At-least-once delivery is assumed; consumers commit inbox state with aggregate mutation.
- Ordering exists only within an aggregate/session; cross-aggregate order derives from event time and causal references.
- No distributed transaction spans EHR, communication provider and platform state; sagas persist intent, acknowledgement and reconciliation.
- Retries are deadline-aware, jittered and eligible only when the operation and commit state permit.
- Long-running workflows pin compatible artifact versions or migrate at an explicit safe boundary.

5 Solutioning Approach

5.1 Design Principles

- Deterministic spine; agentic side branches.
- Compile policy before retrieval and tools.
- Make uncertainty a domain state, not prose.
- Prefer rebuildable projections over competing truths.
- Treat every behavior-affecting artifact as a release.
- Design kill switch, rollback and degraded mode before happy-path scale.

5.2 Architecture Method

Start with a decision inventory, classify each decision by reversibility, authority, data class, latency, failure asymmetry, isolation and accountable human, and then test a thin vertical slice across ingress, identity, policy, retrieval, model, tool, external side effect, reconciliation, telemetry and audit. Technology selection follows quality-attribute evidence rather than service popularity.

5.3 Persona Control Plane

Persona	May do	Must not do
Call coordinator	Read approved context, draft and use bounded call controls.	Merge identity, give clinical advice or write unrestricted EHR state.
Visit coordinator	Propose options and request a temporary hold.	Book without policy, confirmation and idempotent transaction.
Gap coordinator	Open outreach tasks and classify responses.	Close a measure without authoritative evidence.
Transition coordinator	Assemble context and propose follow-up tasks.	Change clinical plan or suppress required review.
Referral coordinator	Extract packet and request missing items.	Invent missing facts or close without acknowledgement.
Care-manager copilot	Summarize and draft plan tasks.	Alter goals or clinical instructions autonomously.

5.4 Provider-Independent Orchestration

Domain workflows call capability names such as grounded-summary:v3, outreach-draft:v2 and appointment-rank:v4, never deployment IDs. The AI gateway resolves capability, risk, region, data class, latency and budget to an already evaluated Foundry or Azure Machine Learning deployment. This isolates model churn from durable workflows.

5.5 Build Versus Managed Service

Build as differentiated control	Use managed Azure substrate
Timeline, decision record, autonomy policy, context compiler, tools, feature semantics and evaluation sets.	Identity, private network, model hosting, search engine, speech, storage primitives and monitoring.
Workflow state machines, EHR reconciliation and typed domain errors.	Durable Functions, Service Bus and API Management execution capabilities.
Point-in-time dataset and outcome learning semantics.	Azure Machine Learning, MLflow, ADLS and managed compute.

6 Reference Architecture

PRINCIPAL ARCHITECTURE POSITION

Ingress is normalized before it enters durable workflow. Model reasoning receives a signed, minimum-necessary context. Side effects are executed only by deterministic services through policy-mediated tools and are reconciled with the external source of truth.

CHANNELS → FRONT DOOR/WAF → API MANAGEMENT → CHANNEL ADAPTERS

POSTGRESQL + OUTBOX → EVENT GRID / SERVICE BUS → DURABLE FUNCTIONS

POLICY + CONTEXT COMPILER → FOUNDRY AGENT TASK → APIM MCP TOOL GATEWAY

EHR / SCHEDULING / OUTREACH / REFERRAL → RECEIPT + RECONCILIATION

6.1 Control Flow of a Single Decision

1. Authenticate workload and derive tenant, subject, purpose, channel, consent and trace claims.
2. Deduplicate the envelope; persist source pointer and canonical fingerprint.
3. Load aggregate with expected version; reject stale command.
4. Compile eligible actions, required evidence, autonomy class and permitted tools.
5. Compile exact facts, filtered retrieval, graph paths and features into a signed context manifest.
6. Invoke capability route with deadline, schema and token budget; validate syntax, semantics, grounding and policy.
7. Create durable human interrupt if required; revalidate on callback.
8. Invoke idempotent tool; persist request hash, external receipt and commit state.
9. Reconcile external authority; append decision and mature outcomes without rewriting history.

6.2 Tenancy and Isolation

The default topology is a regional cell with a tenant cohort. Tenant ID is carried in Entra-issued claims, Service Bus sessions, database row-level security, blob paths, search filters, cache keys, feature keys and telemetry dimensions. Dedicated cells support high-sensitivity or high-scale tenants without changing contracts. The control subscription distributes signed configuration but has no PHI read path.

6.3 Compile and Serve Boundary

Compile mode creates immutable releases for model, prompt, policy, feature, corpus, index and agent artifacts. Serve mode resolves only approved compatible releases. Expensive and nondeterministic builds stay outside the online path; rollback is an alias transition to a known artifact set, not an emergency rebuild.

7 Multi-Agent Orchestration Fabric

PRINCIPAL ARCHITECTURE POSITION

Durable Functions owns lifecycle and external-effect semantics; Microsoft Foundry agents perform bounded reasoning. The split is semantic: anything that waits, compensates, mutates external state or must replay belongs to deterministic workflow.

7.1 Runtime Boundary

Belongs in durable workflow	Belongs in bounded agent task
Timers, retries, human waits, state transitions, compensations, side effects and authoritative history.	Context selection within policy, typed proposal, summary, critique and tool-choice recommendation.
Pinned release set, expected aggregate version and reconciliation state.	Step/token/tool/retrieval budget, schema output and explicit abstention.

7.2 Agent Inventory

Agent	Typed output	Maximum authority
Intake normalizer	NormalizedIntent	Proposal only.
Longitudinal synthesizer	LongitudinalSummary	Read-only; no canonical mutation.
Coordination planner	CoordinationPlan	Ranks admissible actions only.
Evidence challenger	DefectReport	May block promotion; cannot execute.
Communication author	OutboundDraft	Draft only.
Referral packet analyst	ReferralPacketAssessment	Cannot invent or close.
Human-review clerk	ReviewBrief	Cannot approve.

7.3 Graph State and Termination

```
AgentRunState { run_id, tenant_id, subject_ref, workflow_instance_id, capability, risk_class, artifact_versions, context_manifest_ref, allowed_tools, budgets, proposals, defects, human_interrupt, trace_id, terminal_state }
```

A supervisor is a finite deterministic graph router, not an unconstrained manager. Parallel read-only nodes merge only commutative fields. Cycle detection hashes (node, validated-state, artifact-set); repetition terminates the run. One schema-repair and one grounding-repair attempt are permitted. Budget exhaustion becomes AGENT_BUDGET_EXHAUSTED, never a partial success.

7.4 Human Interrupt

Review is a durable external event containing review ID, exact artifact hash, evidence manifest, options, reason-code taxonomy, reviewer role, expiry and aggregate version. A callback revalidates consent, policy, state and external availability. Approval of artifact N cannot authorize N+1.

7.5 Autonomy Policy

Autonomy is calculated outside the agent from action class, tenant policy, language scorecard, calibrated confidence, recent override rate, data freshness, identity confidence and dependency health. The only results are AUTO_EXECUTE, DRAFT_FOR_REVIEW, HUMAN_ONLY or PROHIBITED. Live quality feedback can reduce autonomy; increasing it requires reviewed promotion evidence.

7.6 Tool Fabric

Internal tools are versioned MCP-compatible capabilities mediated by API Management. APIM validates managed identity, tenant/purpose claims, capability, schema, rate, payload class and destination. No generic HTTP tool exists. External organizational handoff uses signed task/evidence envelopes through APIs or events; transitive tool authority never crosses the boundary.

7.7 Trajectory Evaluation

Evaluation scores both artifact quality and trajectory correctness: permitted-node precision, forbidden-tool attempts, state-merge determinism, step efficiency, budget compliance, interrupt correctness, duplicate callback recovery, evidence completeness and outcome utility. Acceptable prose produced by an invalid path still fails release.

8 Feature Platform

PRINCIPAL ARCHITECTURE POSITION

Feature governance concerns meaning, time, lineage and use—not merely storage. Online inference and offline training must consume the same declared computation at different as-of times.

8.1 Feature Contract

```
FeatureDefinition { name, entity_keys, event_time, availability_time, source_lineage, transform_digest, freshness_sla, null_semantics, data_class, permitted_purposes, owner, version }
```

8.2 Entity and View Model

Entity/view	Examples	Freshness/use
Patient engagement	Attempt count, channel response, fatigue and preferred time.	Minutes-hours; outreach selection.
Timeline acuity	Recent discharge, unresolved escalation and event density.	Minutes; queue priority, never diagnosis.
Care-gap evidence	Evidence age, source authority and completion confidence.	Event/daily; eligibility.
Referral readiness	Checklist completeness, destination eligibility and aging.	Minutes; routing.
Scheduling fit	Hard eligibility, language/travel constraints and slot utility.	Seconds; candidate ranking.
Model reliability	Route health, abstention and subgroup quality.	Minutes; autonomy reduction.

8.3 Online, Offline and Streaming Semantics

ADLS Gen2 stores versioned offline history. PostgreSQL holds authoritative compact coordination facts. Cosmos DB or Redis serves hot views only where latency justifies it. Event Hubs plus Stream Analytics or Functions compute bounded rolling aggregates; Synapse or Azure Machine Learning jobs build historical transformations. Online values include feature timestamp, materialization timestamp and definition version; stale required features prevent autonomous use.

8.4 Point-in-Time Correctness

A training row joins only values whose availability time is less than or equal to the decision cut-off. Event time alone is insufficient because older events may have become available later. The compiler freezes candidate universe, policy version, corpus cut-off, exposure probability and label maturity, preventing hindsight leakage through retrieval, negative sampling and latest-state joins.

8.5 Monitoring

Signal	Control response
Freshness breach	Demote autonomy or stop dependent capability.
Training-serving skew	Freeze promotion and compare served versus recomputed vector.
Distribution drift	Diagnose source, population or workflow change before retraining.
Null/default drift	Quarantine source and prohibit silent imputation.
Leakage canary	Invalidate dataset and downstream scorecard.
Subgroup coverage/error	Restrict intended use or require review.

9 Model Portfolio

Capability	Model/control	Serving path	Blocking/release evidence
Intent/entity extraction	Compact transformer plus validators	Azure ML online endpoint	F1, calibration, language strata and schema validity.
Identity candidate scoring	Pairwise model plus hard rules	Azure ML endpoint/batch	Precision at merge threshold and sentinel safety set.
Next-best action	LightGBM/LambdaMART after eligibility	Batch and online	NDCG/uplift, calibration, monotonicity and exposure audit.
Propensity/risk	Calibrated classifier	Batch score	Brier/ECE, subgroup error, actionability and drift.
Safety sentinel	Independent ensemble and high-recall rules	Low-latency endpoint	Recall floor, multilingual adversarial suite and burden.
Generated artifact	Foundry model route with structured output	AI gateway	Grounding, omission, contradiction, policy and injection.
Embedding/rerank	Foundry embedding plus evaluated reranker	Azure AI Search	Recall@k, nDCG, deletion, latency and support.

9.1 Ranking in Depth

Ranking starts only after deterministic eligibility creates the candidate universe. Training retains surfaced candidates and exposure probability; unexposed actions are not assumed negative. Labels mature through contacted, scheduled, completed, acknowledged and closed states. Rank score stays distinct from calibrated outcome probability and policy utility. Approved low-risk randomized exploration may reduce position bias; high-risk actions remain non-exploratory.

9.2 Communication Generation

Tenant voice is a versioned style profile with approved exemplars and terminology constraints. Fine-tuning is considered only after prompting and retrieval have measurable limitations and enough adjudicated edits exist. Every draft is checked for recipient, unsupported policy claim, prohibited clinical language, factual evidence, locale, readability and placeholder integrity.

9.3 Safety Sentinel

The sentinel is independent of the authoring model and evaluates inbound and outbound paths. It combines deterministic rules, compact classifiers, language-specific thresholds and explicit UNCERTAIN. Positive or uncertain high-severity results suppress autonomous generation and invoke a pre-approved human workflow with exact evidence spans.

10 Custom Training Pipeline

PRINCIPAL ARCHITECTURE POSITION

Dataset construction is a signed build. Promotion is separate from training. Aggregate accuracy cannot compensate for leakage, poor calibration, unsafe subgroup behavior, unbounded latency or absent rollback.

10.1 Pipeline Stages

1. Register intended use, decision owner and threat model.
2. Resolve rights-approved source manifests and retention constraints.
3. Tokenize/de-identify and verify recall where training does not require identifiers.
4. Compile point-in-time features, retrieval corpus and candidate universe.
5. Create subject/episode/tenant-grouped temporal splits.
6. Train and calibrate with pinned environment, seed and compute.
7. Evaluate robustness, subgroup performance, leakage, privacy, latency, security and cost.
8. Register model, environment, manifest, model card and scorecard in MLflow/Azure ML.
9. Deploy shadow, then canary behind a capability route.
10. Promote after online parity and rollback rehearsal.

10.2 Dataset and Tenant Governance

Eligibility is evaluated per record using agreement, purpose, retention, de-identification status and withdrawal state. Shared training excludes tenant data by default. Deletion traverses source → derived rows → embeddings → features → datasets → adapters → evaluation caches. An affected model remains under deletion remediation until replacement or documented unlearning treatment.

10.3 Evaluation Hierarchy

Level	Evidence
Component	Schema, accuracy, calibration, robustness and latency.
Retrieval	Recall, ranking, citation support, contradiction and deletion completeness.
Agent trajectory	Node/tool correctness, budgets, termination and recovery.
Workflow	State transition, idempotency, reconciliation and human interrupt.
Prospective shadow	Population, workflow and operator behavior without side effects.
Canary	Bounded live exposure with explicit rollback threshold.
Outcome	Incremental verified operational result plus guardrails.

10.4 Promotion Gates

Data rights, leakage tests, task quality, safety suites, subgroup constraints, artifact supply-chain scan, p95/p99 latency, quota, fallback, cost, owner, expiry and rollback target must all pass. Emergency safety rules ship through the deterministic policy path; they do not bypass model lifecycle controls.

11 AI Control Plane

11.1 Model Gateway

API Management is the only model-egress path. It validates workload identity and purpose, enforces data-class route eligibility, applies rate/token/cost quotas, injects immutable trace and decision IDs, records usage, and blocks direct endpoint access through network policy. Standard logs never capture raw PHI prompts; the call ledger references encrypted restricted payload storage only when retention is permitted.

11.2 Capability Router

```
candidate.where(capability).where(schema-compatible).where(data-class-approved).where(region-allowed).where(scorecard=APPROVED).where(not-expired).order-by(risk-adjusted-quality, deadline-fit, marginal-cost).first-or-abstain()
```

Fallback is an evaluated route, not arbitrary model substitution. When no candidate satisfies the contract, the gateway returns NO_APPROVED_MODEL_ROUTE and the workflow selects a deterministic template or human path.

11.3 Context Compiler

The compiler turns broad data into the minimum admissible context for one decision. It evaluates tenant and purpose policy, selects a longitudinal window, obtains structured facts, performs ACL-filtered hybrid retrieval, expands approved graph paths, reserves contradiction budget, tokenizes identifiers, and emits a signed manifest. System policy, structured facts, retrieved text and user content remain separate channels.

```
ContextManifest { tenant_id, subject_ref, purpose, as_of, evidence[source, interval, authority, hash], retrieval[index, filters, ranks, reranker], omissions, redactions, token_allocation, policy_digest, manifest_hash }
```

12 Knowledge and Retrieval

PRINCIPAL ARCHITECTURE POSITION

Search is an evidence-discovery projection, never authority. Raw EHR payloads are not indiscriminately embedded. Structured facts stay canonical; indexed chunks retain source identity, subject, tenant, time, sensitivity, purpose and deletion lineage.

12.1 Knowledge Domains

Indexes are separated by authority and cadence: tenant policy, scheduling/provider facts, referral requirements, program content, released templates, patient-specific released evidence and operational playbooks. Index aliases support atomic promotion and rollback. High-sensitivity generation has no shared semantic cache.

12.2 Hybrid Retrieval Pipeline

1. Compile tenant, subject, purpose, time and sensitivity filters before similarity.
2. Normalize query and identify exact codes, names and temporal constraints.
3. Execute BM25 and vector retrieval in parallel within the authorized set.
4. Fuse channels while preserving raw ranks and scores.
5. Apply semantic reranking, source diversity and contradiction quotas.
6. Hydrate graph paths and structured facts from authoritative services.
7. Reject stale, superseded, unsupported or inaccessible evidence.
8. Pack by authority and token budget; emit claim-to-source map.

12.3 Embedding and Index Lifecycle

Ingestion performs malware scan, normalization, OCR, structural parsing, section-aware chunking, PHI classification, authorization, embedding and index validation. Chunks retain parent offsets and content hash. Embedding changes create a new index; alias movement requires retrieval regression, support, deletion and latency tests. Vector dimensionality is never changed in place.

12.4 GraphRAG

The graph projects Patient, Episode, Interaction, Appointment, Referral, Task, Evidence, Policy, Provider, Location and Decision artifacts. Queries use reviewed parameterized path templates with hop, row and time budgets. Every edge hydrates to canonical evidence. A model may narrate a retrieved path but may not create predicates, treat similarity as fact, or use graph centrality as authority.

13 Speech and Multilingual Intelligence

13.1 Voice Pipeline

Azure Communication Services terminates calling and Azure AI Speech supplies streaming recognition and synthesis. Session state records codec, language hypothesis, diarization, transcript sequence, consent, identity confidence, barge-in epoch and escalation. Partial transcripts may trigger speculative reads, but no side effect uses an unstable partial. Barge-in cancels synthesis using an epoch token so stale audio cannot play.

Failure	Control
Entity mistranscription	Confirm names, dates and identifiers; use confidence spans and domain phrase lists.
Language/code switch	Maintain rolling language hypothesis; route to bilingual human below threshold.
Stale synthesized response	Playback epoch, cancellation token and deterministic high-risk templates.
Reconnect/duplicate media event	Session sequence and idempotent media command.
Recording consent/retention	Evaluate before capture; separate encrypted object and deletion lineage.

13.2 Multilingual Control

Language support is capability-specific. ASR, intent, safety, translation, generation and TTS each require a scorecard. If direct-language generation is not approved, the platform may create a staff-reviewed translated artifact while preserving original text and alignment spans. Low-resource languages remain human-only for consequential actions until safety recall and adequacy thresholds pass.

14 LLMOps, Observability and AI SRE

14.1 Artifact Inventory

Models, deployments, datasets, features, prompts, schemas, agents, tools, router policies, search indexes, embeddings, rule bundles, red-team corpora and evaluations are first-class artifacts. Every artifact carries owner, intended use, digest, dependency graph, compatibility, approval, expiry, effective interval and rollback target. Latest is forbidden in decision records.

14.2 Tracing Model

OpenTelemetry spans edge request, workflow instance, agent node, retrieval channel, model call, validator, tool authorization, external call, database transaction and reconciliation. PHI is excluded from normal attributes. Critical safety, policy-denial and cross-tenant events are never sampled away.

14.3 AI SLOs

SLO	Error budget response
Decision trace completeness	Block release and repair evidence pipeline.
Grounded field support	Demote capability or increase review.
Hard-policy integrity	Fleet/intent kill switch on any bypass.
Retrieval sufficiency	Rollback corpus, index or router.
Exactly-once external effect	Stop tool and reconcile affected decisions.
Calibration	Disable confidence-based autonomy.
Cost per verified workflow	Optimize route/context while preserving controls.

14.4 Incident Classes

AI incidents include cross-tenant disclosure, unauthorized side effect, fabricated evidence, systematic unsupported claim, unsafe language handling, corrupted lineage, evaluation leakage, runaway agent loop and denial-of-wallet. Each class defines detection, kill-switch scope, evidence preservation, affected-decision query, owner, remediation gate and replay requirement.

15 Resilience, Disaster Recovery and Continuity

PRINCIPAL ARCHITECTURE POSITION

Recovery is not complete when endpoints return 200. It is complete when workflow authority, decision lineage, external effects and reconciliation return to a known consistent state.

15.1 RTO/RPO by Capability

Tier	Capability	Continuity
T0	Identity, safety policy, kill switch and audit intake	Paired-region replication; fail closed.
T1	Active voice and urgent transitions	Admission/callback and deterministic templates.
T2	Timeline/read models and work queues	Geo-replicated state plus event replay.
T3	Interactive retrieval and agents	Approved regional routes; lexical/template fallback.
T4	Training, batch analytics and index build	Restore from signed ADLS manifests.

15.2 Regional Cell Topology

Cells use region-local data paths and private endpoints. Front Door routes only after readiness and residency checks. A write epoch fences the old region and establishes one active workflow authority per tenant. Service Bus namespace aliasing does not substitute for message-state recovery; queue reconstruction is validated from outbox and audit history.

15.3 Recovery Lifecycle

Freeze promotion, stop new side effects, capture watermarks, fence old writers, activate signed configuration, verify identity/network/secrets, restore state, reconstruct queues, reconcile uncertain commits, run synthetic probes, and reopen action classes in risk order. Long-running workflows resume only when pinned artifact sets are available and compatible.

15.4 Graceful Degradation

Dependency	Preserve	Disable/Degrade
Foundry model route	Deterministic workflow, template and human queue.	Free-form generation and agent promotion.
Azure AI Search	Canonical fact lookup and released cache.	RAG requiring missing evidence.
EHR connector	Internal task and queued intent.	Assumption of external completion.
Speech synthesis	Text or live transfer where allowed.	Automated spoken response.
Feature online store	Static eligibility and fresh last-known values.	Ranking/autonomy dependent on stale features.

16 Core Tech Stack

PRINCIPAL ARCHITECTURE POSITION

Technology follows authority, isolation, recovery and operability requirements. Preview capability can be isolated behind an adapter but cannot be the sole control for tenancy, safety or audit without explicit risk acceptance.

Layer	Azure technology	Architectural use
Edge	Front Door Premium, WAF, API Management	Global admission, API governance and mediated AI/tool surface.
Channels	Communication Services, Event Grid	Calls, SMS and event normalization.
Workflow	Durable Functions, Service Bus	Long waits, ordered sessions, retry, DLQ and compensation.
Agents/models	Microsoft Foundry Agent Service and model catalog	Bounded reasoning and approved model routes.
Custom ML	Azure Machine Learning and MLflow	Reproducible pipelines, registry and online/batch endpoints.
Search	Azure AI Search	Filtered hybrid retrieval and semantic reranking.
Health data	Azure Health Data Services FHIR service	Managed interoperable resource boundary.
Operational data	Azure Database for PostgreSQL, Cosmos DB, Redis	ACID aggregates, projections and nonauthoritative cache.
Data lake	ADLS Gen2 and Synapse	Immutable history, point-in-time data and analytics.
Security	Entra ID, Managed Identity, Key Vault, Private Link, Defender, Sentinel	Zero trust, secrets, controlled egress and threat response.
Observability	Azure Monitor, Application Insights, Log Analytics, OpenTelemetry	Cross-plane AI and workflow telemetry.
Delivery	Bicep, ACR, Azure DevOps/GitHub Actions, AKS/Container Apps	Signed, reproducible and progressively delivered infrastructure.

16.1 Service Boundary Decisions

Durable Functions owns persisted orchestration; Foundry agents are tasks. PostgreSQL owns transactional aggregates and outbox; Cosmos DB is selected only for access patterns that justify distributed documents or projections. Azure AI Search is rebuildable. AKS is reserved for custom scheduling/sidecars or serving control; Container Apps and Functions reduce platform burden for simpler services.

16.2 Deliberately Rejected Patterns

Rejected pattern	Reason
One autonomous agent end to end	Cannot guarantee replay, least privilege, termination or external-effect semantics.
Vector store as source of truth	Similarity has no transaction, temporal authority or admissibility.
Prompt-only safety/tenancy	Instructions are not authorization.
One frontier model for every task	Unnecessary cost/latency and correlated failure.
Active-active writes everywhere	Split-brain complexity exceeds need.
Embed all EHR content	Expands PHI, deletion burden and irrelevant retrieval.
Fine-tune before retrieval/evaluation maturity	Weights cannot repair absent evidence contracts or labels.

17 Data Architecture and Interoperability

17.1 Operational Stores

Store	Authoritative content	Semantics
PostgreSQL	Aggregates, approvals, idempotency, outbox and decision headers.	ACID, optimistic versioning, RLS and bitemporal fields.
AHDS FHIR	Selected interoperable healthcare resources.	FHIR versioning and scoped access.
ADLS Gen2	Raw immutable payloads, history, features and datasets.	Manifested, partitioned and rights-aware.
Azure AI Search	Retrieval projections and embeddings.	Versioned aliases and rebuildable indexes.
Cosmos DB	High-scale timeline/read/graph projections.	Tenant/subject partition and chosen consistency.
Redis	Ephemeral locks, counters and non-PHI cache.	Fencing; no patient semantic cache.
Audit store	Security events and restricted decision lineage.	Append-only export and retention.

17.2 Canonical AI Decision Record

```
CareCoordinationDecisionRecord { decision_id, tenant_id, subject_ref, episode_id, workflow_id, trace_id, type, risk, autonomy,
state_version, decided_at, input_manifest, context_manifest, feature_vector,
artifacts[model,prompt,schema,agent,tool,policy,index,embedding], proposal_hash, calibrated_confidence, uncertainty, validators,
human_disposition, execution[idempotency,request_hash,receipt,commit_state], outcomes, supersedes, retention, integrity_hash }
```

17.3 FHIR and EHR Boundary

The interoperability model uses Patient, Practitioner, PractitionerRole, Organization, Location, Schedule, Slot, Appointment, Task, ServiceRequest, Communication, Encounter, EpisodeOfCare, CarePlan, Goal, Observation, QuestionnaireResponse, Consent and Provenance. Vendor adapters retain source IDs and versions. Writes are conditional/version-aware where supported. Event-driven and scheduled reconciliation compare intended, acknowledged and observed state; discrepancies create work rather than silent overwrite.

18 Canonical APIs

Surface	Endpoint family	Invariant
AI gateway	POST /v1/capabilities/{name}:invoke	Purpose, data class, deadline, budget, schema and trace required.
Agents	POST /v1/agent-runs; /{id}:resume	Async, pinned artifacts, callback binds hash.
Context	POST /v1/context-manifests:compile	As-of, subject, purpose and evidence policy required.
Features	POST /v1/features:online-get; :historical-materialize	Versioned features and as-of required.
Tools	POST /v1/tools/{capability}:invoke	Managed identity, capability, idempotency and expected state version.
Reviews	GET /v1/reviews; POST /{id}:decide	Role, reason, exact proposal hash and expiry check.
Routes	POST /v1/model-routes:promote; :rollback	Scorecard, compatibility, approvals and rollback target.
Governance	POST /v1/kill-switches	Fleet/cell/tenant/intent/capability/tool/route scope.

18.1 Typed Error Semantics

Domain errors include IDENTITY_AMBIGUOUS, CONSENT_DENIED, EVIDENCE_INSUFFICIENT, CONTEXT_STALE, POLICY_DENIED, MODEL_ROUTE_UNAVAILABLE, SCHEMA_INVALID, HUMAN_REVIEW_REQUIRED, TOOL_COMMIT_UNKNOWN, EXTERNAL_STATE_CONFLICT, FEATURE_STALE and TENANT_BOUNDARY_VIOLATION. Each defines retry eligibility, workflow transition, audit severity, user-safe message and recovery authority. An agent cannot reinterpret a denial.

19 Trade-Offs and Decisions

PRINCIPAL ARCHITECTURE POSITION

The most consequential decisions preserve authority and recoverability even when they add integration work. Principal architecture optimizes the lifecycle of a verified decision—not the apparent simplicity of a model demo.

19.1 Hardest Call: Latency Versus Control

Safety, consent, identity and recipient validation remain on the critical path. Latency is recovered through speculative read-only work, streaming overlap, compact sentinels and deterministic acknowledgement—not by sending first and validating later.

19.2 Hardest Call: Managed Agents Versus Durable Authority

Foundry Agent Service accelerates bounded reasoning, but durable workflow, business state and external effects stay explicit. This produces more visible contracts and adapter work while eliminating hidden agent loops as state authority.

19.3 Hardest Call: Personalization Versus Isolation

Tenant style and policy are retrieval/configuration artifacts first. Shared training is opt-in and de-identified; tenant adapters remain isolated and deletable. Lower pooled-data volume is accepted to make rights, deletion and leakage boundaries enforceable.

DECISION-ENGINEERING RULE

Every architectural decision must identify the authoritative state, enforcement point, telemetry, kill switch, rollback route, accountable human, failure mode and reconsideration trigger. None may be inferred from model-generated prose.

20 Trust, Safety and Cost Governance

20.1 Threat Model

Threat	Primary controls
Prompt injection through message, PDF or search chunk	Instruction/data separation, read-only evidence agent, independent tool policy and outbound verification.
Cross-tenant disclosure	Tenant claim enforced at identity, network, RLS, search, cache, feature and telemetry layers.
Excessive agency	Typed allow-listed tools, parameter policy, expected state version, idempotency and review.
Evidence poisoning	Approved source, content hash, provenance, quarantine and corpus release.
Model/data leakage	Purpose grant, de-identification, restricted logs and extraction/privacy tests.
Denial of wallet	Step, token, retrieval, graph, tool, deadline and monetary budgets.
Supply chain compromise	Pinned digest, signature, SBOM, vulnerability scan and registry approval.

20.2 Injection and Exfiltration Defence

Normalize hidden text and confusables, classify untrusted content, keep instruction hierarchy outside retrieval, separate read-only from tool-capable nodes, authorize tools with short-lived tenant/purpose/capability claims, validate recipients and parameters deterministically, verify grounding after generation, and automatically reduce autonomy on attack evidence. No generic outbound HTTP tool exists; recipient identity comes from verified canonical state, never generated text.

20.3 Tenant-Level Network Enforcement

Subscriptions and regional cells contain administrative blast radius. Private Link removes public data-plane exposure. Workload identity and network policy constrain service paths. APIM and Azure Firewall centralize approved ingress/egress. Automated probes test both allowed connectivity and forbidden cross-cell/public routes.

20.4 Clinical Boundary

The platform coordinates, reminds, summarizes and routes under approved content. It does not diagnose, prescribe, change treatment, interpret results independently to a patient, or replace licensed judgement. Break-glass access requires reason, alert and review. Minimum-necessary context, encryption, immutable access audit and deletion lineage are technical controls.

20.5 Cost Governance

A decision-level ledger attributes model tokens, embeddings, search, speech, workflow, storage, tools and review to tenant, capability, workflow and verified outcome. The router optimizes within a quality floor. Smaller models, compression, caching and batch inference are allowed only after task-specific evaluation. Safety and access control are never degradation knobs.

20.6 Release Vetoes

Cross-tenant exposure, forbidden action bypass, untraceable decision, missing rollback, evaluation leakage, critical safety regression, unresolved uncertain commits, non-replayable migration or undeletable tenant-derived artifact blocks release. A noncritical exception must be owned, time-bound, monitored and paired with compensating control.

21 Authority

21.1 Decision Rights

Architecture owns cross-cutting contracts, authority boundaries, quality attributes, isolation, AI lifecycle gates, resilience tiering and exception process. Product owns intent and user experience; clinical governance owns clinical boundaries; security owns security policy; data governance owns rights/retention; service teams own implementation within guardrails. No owner can unilaterally weaken another owner's hard control.

21.2 Architecture Veto

A veto cites the violated invariant, evidence, affected decisions, safe alternative and exit criteria. It is recorded as an ADR or risk item, not informal opinion. The decision reopens when telemetry, threat evidence, service capability, regulation or workload invalidates the original assumptions.

21.3 Authority Boundary

The Principal AI Architect does not approve clinical content, accept legal risk, or attest compliance. The role ensures those accountable authorities are explicit, represented in software and impossible for agent composition to bypass.

22 Leadership

22.1 Team Topology

The delivery model separates AI control plane, workflow/integration, data/feature, model factory, security/SRE and domain squads. Enabling teams publish paved roads: agent template, tool SDK, decision-record library, evaluation harness, feature contract, EHR connector framework and observability conventions. Domain squads own their state machines and SLOs.

22.2 Technical Governance

Architecture review requires executable scenarios and failure demonstrations: timeout behavior, idempotency, artifact pinning, tenant denial, evaluation evidence and rollback. Reference implementations become conformance tests. Principal attention concentrates on irreversible contracts and cross-team coupling; local implementation remains with owning engineers.

22.3 Programme Sequence

Delivery proceeds through identity/network/audit; canonical contracts/event spine; durable workflow/reconciliation; retrieval/feature foundations; predictive models; bounded agents; autonomy; learning loop. Each phase exits with evidence. Feature concurrency is allowed only when required control-plane dependencies are production-ready.

23 Business Outcome

23.1 Measurement

No fabricated result is asserted. Measures link decision exposure, mature outcome and guardrail: time to successful contact, time to scheduled follow-up, referral closure time, completed tasks, no-show reduction, staff handling time, override rate, opt-out, duplicate action, reconciliation debt and cost per verified outcome.

23.2 Causal Attribution

Where ethical and operationally feasible, stepped-wedge rollout, randomized message timing, shadow ranking and matched controls estimate incremental effect. Delivery or booking is not success when attendance, closure or completed follow-up is the intended outcome. Guardrails prevent optimization that shifts burden to patients or staff.

23.3 Platform Economics

Economics are measured per workflow lifecycle. The ledger separates fixed cell cost, marginal model/search/speech cost, review cost and external-channel cost. A cheaper inference route that increases review, error or rework can be more expensive end to end, so routing optimizes quality-adjusted cost.

24 Architectural Impact

24.1 Reusable Patterns

- Durable workflow / bounded agent split.
- Capability-based model routing with evaluated fallback.
- Signed context manifest and canonical AI decision record.
- Policy before retrieval and policy before tools.
- Point-in-time features and datasets.
- Uncertain-commit reconciliation.
- Retractable autonomy based on live evidence.
- Decision-level cost, trace and artifact lineage.

24.2 Principal Acceptance Criterion

The platform is acceptable only when every consequential action reconstructs from admissible evidence, exact artifacts and workflow history; every probabilistic component can fail without corrupting canonical state; every uncertain external effect can reconcile; tenant and patient boundaries live outside prompts; and operators can stop, degrade, recover and roll back the system without trusting model-generated explanations.

Technical Decision Compendium

The following packets convert architectural intent into enforceable operating decisions. They are deliberately explicit about why, where, when, mechanics, failure behavior, validation and reopening conditions.

ADR-001 · Durable workflow owns lifecycle; agents are bounded tasks

DECISION	Durable workflow owns lifecycle; agents are bounded tasks
WHY	Long waits, replay, human interrupts and external effects require deterministic state. Foundry agents are optimized for reasoning, not transaction authority.
WHERE	Durable Functions orchestrations and domain workflow services; agent calls occur as activities.
WHEN	Any workflow crossing process lifetime, external API, timer, retry or human review.
EXECUTION	Orchestrator persists state and artifact pins; activities perform I/O; agent returns typed proposal; deterministic code validates and transitions.
FAILURE / DEGRADED MODE	If agent runtime fails, retry only within deadline, then route to evaluated fallback or human queue. Canonical state remains unchanged.
VALIDATION EVIDENCE	Replay determinism, duplicate activity, callback race, timeout and regional recovery tests.
RECONSIDERATION TRIGGER	Reopen if managed agent runtime proves equivalent persisted state semantics, migration control and external-effect recovery.

ADR-002 · Capability-based AI gateway

DECISION	Capability-based AI gateway
WHY	Binding workflows to deployment IDs couples domain behavior to model churn and bypasses policy/cost controls.
WHERE	API Management AI gateway and private Foundry/Azure ML endpoints.
WHEN	Every model, embedding and reranking call.
EXECUTION	Resolve capability, schema, risk, data class, region, deadline and budget against approved scorecards.
FAILURE / DEGRADED MODE	No eligible route returns a typed denial; workflow selects template or review, never arbitrary substitution.
VALIDATION EVIDENCE	Route unit tests, scorecard compatibility, throttling, quota exhaustion and rollback drills.
RECONSIDERATION TRIGGER	Reopen routing objectives when measured quality/latency/cost frontier changes.

ADR-003 · Compile minimum-necessary signed context

DECISION	Compile minimum-necessary signed context
WHY	Unbounded prompts increase leakage, stale facts, cost and audit ambiguity.
WHERE	Context compiler before every generative or agent task.
WHEN	When a decision needs structured facts, retrieval, graph or longitudinal history.
EXECUTION	Policy selects evidence; compiler records source/time/authority/hash, omissions and token allocation; manifest is signed.
FAILURE / DEGRADED MODE	Insufficient or stale required evidence forces abstention or review.
VALIDATION EVIDENCE	Claim support, temporal cut-off, ACL, redaction, prompt-injection and manifest replay tests.
RECONSIDERATION TRIGGER	Reopen packing strategy when model context characteristics or evidence domains materially change.

ADR-004 · Policy precedes retrieval

DECISION	Policy precedes retrieval
WHY	Post-retrieval filtering exposes unauthorized content to memory, logs and caches.
WHERE	Search query construction and graph hydration.
WHEN	Every tenant/subject/purpose-scoped retrieval.
EXECUTION	Compile hard filters before BM25/vector execution; hydrate only authorized IDs.
FAILURE / DEGRADED MODE	Denied or ambiguous scope produces no result, not best-effort filtering.
VALIDATION EVIDENCE	Canary documents, cross-tenant negative tests and deletion verification.
RECONSIDERATION TRIGGER	Never reopen the ordering; only optimize implementation.

ADR-005 · Deterministic authorization for every tool

DECISION	Deterministic authorization for every tool
WHY	A language model is not a policy decision point.
WHERE	APIM tool gateway and downstream domain service.
WHEN	Any mutation or sensitive read.
EXECUTION	Validate managed identity, tenant, purpose, capability, schema, expected version, recipient, rate and idempotency.
FAILURE / DEGRADED MODE	Deny before invocation and emit immutable reason; the agent cannot retry with expanded scope.
VALIDATION EVIDENCE	Fuzz parameters, privilege escalation, replay and confused-deputy tests.
RECONSIDERATION TRIGGER	Reopen allowed capability only through threat review and policy promotion.

ADR-006 · No generic HTTP tool

DECISION	No generic HTTP tool
WHY	Generic egress creates an unbounded exfiltration and side-effect surface.
WHERE	Agent tool registry, network policy and firewall.
WHEN	All agentic execution.
EXECUTION	Expose narrow named operations with typed request/response and fixed destinations.
FAILURE / DEGRADED MODE	Unknown tool or destination is blocked and alerted.
VALIDATION EVIDENCE	DNS/egress probes and generated URL attacks.
RECONSIDERATION TRIGGER	Reopen only for a dedicated sandbox with no PHI, no credentials and explicit egress broker.

ADR-007 · Outbox/inbox and uncertain-commit reconciliation

DECISION	Outbox/inbox and uncertain-commit reconciliation
WHY	At-least-once delivery and timeout after commit make naive retry unsafe.
WHERE	Transactional services, Service Bus consumers and connector services.
WHEN	Every event publication and external mutation.
EXECUTION	Commit aggregate plus outbox; consume with inbox; external request carries idempotency key; unknown commit is queried before retry.
FAILURE / DEGRADED MODE	Hold workflow in COMMIT_UNCERTAIN and page/queue by risk and age.
VALIDATION EVIDENCE	Crash-point injection before/after commit, lost acknowledgement and duplicate replay.
RECONSIDERATION TRIGGER	Reopen per connector only when authoritative API guarantees equivalent idempotency.

ADR-008 · One writer per aggregate

DECISION	One writer per aggregate
WHY	Concurrent agent branches must not create competing canonical states.
WHERE	PostgreSQL domain services.
WHEN	Mutations to episode, referral, appointment, outreach and decision aggregates.
EXECUTION	Optimistic version check; commands state expected version; conflicts trigger recomputation.
FAILURE / DEGRADED MODE	Stale mutation fails with EXTERNAL_STATE_CONFLICT.
VALIDATION EVIDENCE	Concurrent command, retry and saga compensation tests.
RECONSIDERATION TRIGGER	Reopen granularity if aggregate contention violates measured SLO.

ADR-009 · Bitemporal longitudinal facts

DECISION	Bitemporal longitudinal facts
WHY	Late and corrected events make current-value storage non-replayable.
WHERE	Timeline and longitudinal projection.
WHEN	Facts whose effective and known times differ.
EXECUTION	Store valid interval and system interval; corrections append new versions; decisions use as-of cut-off.
FAILURE / DEGRADED MODE	Projection rebuilds affected interval and marks dependent decisions superseded.
VALIDATION EVIDENCE	Late event, correction, deletion and replay parity tests.
RECONSIDERATION TRIGGER	Reopen physical implementation, never temporal semantics.

ADR-010 · Point-in-time feature and dataset compiler

DECISION	Point-in-time feature and dataset compiler
WHY	Latest-value joins and latest corpus leak future information.
WHERE	Feature platform and training pipeline.
WHEN	Training, backtesting and retrospective evaluation.
EXECUTION	Join on availability time; freeze candidate universe, policy, corpus, exposure and maturity.
FAILURE / DEGRADED MODE	Compiler rejects records with unresolved time/lineage.
VALIDATION EVIDENCE	Leakage canaries, temporal split audit and online/offline parity.
RECONSIDERATION TRIGGER	Reopen implementation when native platform capabilities preserve the same contract.

ADR-011 · Search is a rebuildable projection

DECISION	Search is a rebuildable projection
WHY	Indexes optimize discovery but cannot provide transaction or authority.
WHERE	Azure AI Search.
WHEN	Every retrieval use.
EXECUTION	Canonical source emits versioned projection; alias points to promoted index.
FAILURE / DEGRADED MODE	Rollback alias or rebuild from manifest.
VALIDATION EVIDENCE	Rebuild parity, stale deletion, alias swap and retrieval regression.
RECONSIDERATION TRIGGER	Never promote search to source of truth.

ADR-012 · Structured facts are not embedded by default

DECISION	Structured facts are not embedded by default
WHY	Embedding exact, sensitive facts expands exposure and weakens precision.
WHERE	Ingestion and context compiler.
WHEN	EHR fields, identity, consent, dates, codes and current statuses.
EXECUTION	Read from authoritative service and render typed facts; embed only approved narrative evidence.
FAILURE / DEGRADED MODE	Unavailable exact fact makes dependent decision ineligible.
VALIDATION EVIDENCE	Source comparison, freshness and deletion tests.
RECONSIDERATION TRIGGER	Reopen a field only with use case, risk and retrieval evidence.

ADR-013 · Hybrid retrieval plus semantic reranking

DECISION	Hybrid retrieval plus semantic reranking
WHY	Vector-only search loses exact codes/names; lexical-only misses paraphrase.
WHERE	Azure AI Search query service.
WHEN	Evidence discovery over approved corpus.
EXECUTION	Run lexical and vector channels, fuse, rerank, enforce source diversity and contradiction budget.
FAILURE / DEGRADED MODE	If reranker unavailable, use evaluated fusion route or abstain by risk.
VALIDATION EVIDENCE	Recall@k, nDCG, entailment, contradiction, latency and deletion.
RECONSIDERATION TRIGGER	Reopen weights/model on corpus or query-distribution drift.

ADR-014 · GraphRAG uses reviewed path templates

DECISION	GraphRAG uses reviewed path templates
WHY	Free model-authored graph queries are difficult to bound and authorize.
WHERE	Graph projection/query API.
WHEN	Multi-hop relationship evidence.
EXECUTION	Select named parameterized template; enforce hop/row/time budget; hydrate each edge to evidence.
FAILURE / DEGRADED MODE	Unknown template/edge authorization fails closed.
VALIDATION EVIDENCE	Path correctness, fan-out, injection, stale edge and evidence hydration.
RECONSIDERATION TRIGGER	Reopen templates through domain and security review.

ADR-015 · Independent safety sentinel

DECISION	Independent safety sentinel
WHY	The same generator should not author and certify its own output.
WHERE	Pre/post-generation safety services.
WHEN	Inbound high-risk detection and outbound validation.
EXECUTION	Use rules plus compact classifiers with language scorecards; uncertain high-severity escalates.
FAILURE / DEGRADED MODE	Suppress automation and route human process.
VALIDATION EVIDENCE	Recall floor, adversarial multilingual suite, workload burden and drift.
RECONSIDERATION TRIGGER	Reopen threshold only with reviewed scorecard and operational-capacity analysis.

ADR-016 · Calibrated confidence controls autonomy

DECISION	Calibrated confidence controls autonomy
WHY	Raw logits and model self-confidence are not probabilities.
WHERE	Model scorecard and policy engine.
WHEN	Confidence-based AUTO_EXECUTE decisions.
EXECUTION	Calibrate on representative set; monitor ECE/Brier; combine with policy, freshness and route health.
FAILURE / DEGRADED MODE	Calibration breach disables confidence-based automation.
VALIDATION EVIDENCE	Reliability diagrams, subgroup calibration and shift tests.
RECONSIDERATION TRIGGER	Reopen calibrator on material distribution change.

ADR-017 · Explicit abstention is a successful terminal result

DECISION	Explicit abstention is a successful terminal result
WHY	Forcing a guess hides uncertainty and increases harmful action.
WHERE	Schemas, workflows and UX.
WHEN	Missing evidence, low confidence, conflicting sources or unsupported language.
EXECUTION	Typed abstention includes reason, missing evidence and next safe path.
FAILURE / DEGRADED MODE	Create clarification or review rather than retry loop.
VALIDATION EVIDENCE	Abstention precision, burden and outcome utility.
RECONSIDERATION TRIGGER	Reopen thresholds based on asymmetric cost and operator capacity.

ADR-018 · Human approval binds an exact artifact

DECISION	Human approval binds an exact artifact
WHY	Approval cannot authorize unseen regeneration.
WHERE	Review API and workflow resume.
WHEN	Any draft/plan requiring human authority.
EXECUTION	Store proposal hash, evidence manifest, version, reason options and expiry; callback revalidates.
FAILURE / DEGRADED MODE	State/consent/policy change invalidates approval.
VALIDATION EVIDENCE	Stale approval, duplicate callback and changed-recipient tests.
RECONSIDERATION TRIGGER	Never weaken binding; optimize review UX only.

ADR-019 · Autonomy is retractable

DECISION	Autonomy is retractable
WHY	Quality can degrade after release due to drift, attack or dependency behavior.
WHERE	Policy engine and kill-switch plane.
WHEN	Every autonomous action class.
EXECUTION	Compute autonomy from current scorecard, SLOs, override, drift, freshness and security signals.
FAILURE / DEGRADED MODE	Automatically demote to review or human-only; promotion remains deliberate.
VALIDATION EVIDENCE	Chaos/drift drills and kill-switch latency.
RECONSIDERATION TRIGGER	Reopen promotion criteria with governance approval.

ADR-020 · Feature definitions are versioned contracts

DECISION	Feature definitions are versioned contracts
WHY	A column name does not encode time, lineage, null or permitted use.
WHERE	Feature registry.
WHEN	Online/offline feature consumption.
EXECUTION	Declare entity, event/availability time, transform, freshness, null, class and purpose.
FAILURE / DEGRADED MODE	Unknown or stale required feature blocks dependent autonomy.
VALIDATION EVIDENCE	Contract conformance and serving parity.
RECONSIDERATION TRIGGER	Reopen definition as a new version, never mutate historical meaning.

ADR-021 · Exposure logging for ranking

DECISION	Exposure logging for ranking
WHY	Observed outcomes depend on which options were shown.
WHERE	Ranking service and decision record.
WHEN	Every recommendation list.
EXECUTION	Record candidate universe, rank, score, policy exclusions, exposure probability and selected action.
FAILURE / DEGRADED MODE	Missing exposure excludes sample from causal/ranking label construction.
VALIDATION EVIDENCE	Propensity integrity and candidate replay.
RECONSIDERATION TRIGGER	Reopen exploration policy with ethics/domain approval.

ADR-022 · Labels mature and never rewrite decisions

DECISION	Labels mature and never rewrite decisions
WHY	Operational truth arrives over days or months.
WHERE	Outcome store and training compiler.
WHEN	Training and outcome reporting.
EXECUTION	Append outcome states with observation time and maturity; link to original decision.
FAILURE / DEGRADED MODE	Late corrections supersede labels but preserve original evidence.
VALIDATION EVIDENCE	Maturity-window and correction tests.
RECONSIDERATION TRIGGER	Reopen maturity by endpoint based on empirical delay distribution.

ADR-023 · Artifact pinning for long workflows

DECISION	Artifact pinning for long workflows
WHY	A workflow cannot silently change behavior mid-episode.
WHERE	Workflow instance and registry resolver.
WHEN	Multi-step, multi-day lifecycle.
EXECUTION	Pin compatible artifact set; allow migration only at declared safe point.
FAILURE / DEGRADED MODE	Missing pinned artifact pauses/requires compatibility migration.
VALIDATION EVIDENCE	Rollback/resume and schema evolution tests.
RECONSIDERATION TRIGGER	Reopen pin duration when safety patch policy requires.

ADR-024 · Independent versioning with compatibility matrix

DECISION	Independent versioning with compatibility matrix
WHY	Bundling all AI artifacts slows rollback and obscures causality.
WHERE	Artifact registry and release controller.
WHEN	Every deployment.
EXECUTION	Version model, prompt, schema, agent, policy, index, embedding, feature and tool independently; test combinations.
FAILURE / DEGRADED MODE	Incompatible resolution is denied.
VALIDATION EVIDENCE	Matrix CI and rollback to known set.
RECONSIDERATION TRIGGER	Reopen packaging granularity if operational overhead outweighs isolation.

ADR-025 · Shadow then canary promotion

DECISION	Shadow then canary promotion
WHY	Offline evaluation cannot capture live workflow and population behavior.
WHERE	Gateway route and deployment pipeline.
WHEN	New model/prompt/router/index/agent release.
EXECUTION	Shadow without effects; compare; canary bounded tenants/intents; observe; expand.
FAILURE / DEGRADED MODE	Automatic rollback on predefined SLO/safety breach.
VALIDATION EVIDENCE	Parity, outcome, override, latency and cost.
RECONSIDERATION TRIGGER	Reopen cohort design as traffic/population changes.

ADR-026 · Decision-level observability

DECISION	Decision-level observability
WHY	Service uptime cannot reveal unsupported or untraceable decisions.
WHERE	OpenTelemetry and decision ledger.
WHEN	Every consequential decision.
EXECUTION	Link trace, workflow, model, retrieval, validators, tool and outcome.
FAILURE / DEGRADED MODE	Missing trace/decision completeness blocks autonomous promotion.
VALIDATION EVIDENCE	Trace completeness query and incident reconstruction drill.
RECONSIDERATION TRIGGER	Reopen retention/sampling based on privacy and incident needs.

ADR-027 · PHI excluded from standard telemetry

DECISION	PHI excluded from standard telemetry
WHY	General logs have broad access and long retention.
WHERE	Application Insights and Log Analytics conventions.
WHEN	All services.
EXECUTION	Use IDs, hashes and reason codes; restricted encrypted payload reference only where allowed.
FAILURE / DEGRADED MODE	Detection triggers quarantine, incident and deletion.
VALIDATION EVIDENCE	DLP scans and seeded canaries.
RECONSIDERATION TRIGGER	Reopen only for explicitly governed restricted store.

ADR-028 · Regional cells with one active write epoch

DECISION	Regional cells with one active write epoch
WHY	Global active-active workflow writes create split brain.
WHERE	Regional subscriptions, data and workflow control.
WHEN	Regional deployment and failover.
EXECUTION	Assign tenant to cell; fence old epoch before paired-cell write activation.
FAILURE / DEGRADED MODE	If fencing uncertain, run read-only/queue mode.
VALIDATION EVIDENCE	Partition/failover and stale-writer tests.
RECONSIDERATION TRIGGER	Reopen active-active for proven conflict-free bounded domains only.

ADR-029 · Risk-tiered degradation

DECISION	Risk-tiered degradation
WHY	Uniform availability either compromises controls or over-stops the system.
WHERE	Workflow and dependency-health policy.
WHEN	Dependency failure.
EXECUTION	Map each capability to required dependencies and approved degraded behavior.
FAILURE / DEGRADED MODE	Hard controls fail closed; low-risk work may template/queue.
VALIDATION EVIDENCE	Dependency chaos matrix.
RECONSIDERATION TRIGGER	Reopen tier after new fallback receives evaluation.

ADR-030 · Decision-level FinOps

DECISION	Decision-level FinOps
WHY	Per-resource bills do not reveal expensive workflows or rework.
WHERE	Usage ledger and analytics.
WHEN	All model/search/speech/tool calls.
EXECUTION	Attribute direct cost and review/retry to decision and mature outcome.
FAILURE / DEGRADED MODE	Budget exhaustion routes to approved cheaper path or review.
VALIDATION EVIDENCE	Allocation completeness and denial-of-wallet tests.
RECONSIDERATION TRIGGER	Reopen route objective when quality-adjusted economics change.

Production Verification Matrix

Production readiness is proven through executable evidence across state, AI, security, recovery and operations. The following tests are release inputs rather than post-release aspirations.

Domain	Test	Release criterion
State/replay	Rebuild timeline, queues and read models from canonical events; compare hashes and decision outcomes.	Zero unexplained divergence.
Idempotency	Inject crash before and after internal/external commit; replay messages and callbacks.	No duplicate consequential effect.
Concurrency	Race state change, human approval and connector acknowledgement.	Expected-version semantics preserve authority.
Retrieval	Evaluate ACL, temporal cutoff, contradiction, deletion and evidence support.	All hard isolation/deletion tests pass; scorecard meets release floor.
Agent trajectory	Inject tool denial, loops, malformed output and budget exhaustion.	Bounded termination and correct terminal state.
Safety	Run multilingual, indirect, obfuscated and adversarial inputs.	Approved recall floor and bounded review burden.
Security	Test prompt injection, tool escalation, cross-tenant probes and exfiltration paths.	No privilege expansion or unauthorized data path.
Resilience	Fail model, search, queue, DB, connector and region; recover pinned workflows.	Tier RTO/RPO plus decision-lineage integrity.
Operability	Exercise route rollback, tool kill switch, tenant isolation and affected-decision query.	On-call performs within runbook target.
Economics	Burst with worst-case token/retry/tool patterns.	Quota/backpressure preserves critical lanes and budget.

Final Principal Architecture Position

The defining property of this platform is not that it invokes generative models. It is that probabilistic components operate inside a deterministic, policy-mediated, temporally correct and recoverable decision system. Longitudinal state is replayable; evidence is admissible and attributable; agents are bounded; tools are least-privileged; human authority is explicit; external effects reconcile; artifacts roll back independently; and autonomy is continuously earned through production evidence.

ACCEPTANCE STATEMENT

No capability is production-complete until its evidence path, authority boundary, failure semantics, observability, rollback, kill switch, deletion lineage, recovery behavior and accountable owner are as concrete as its happy-path model invocation.